

String Theory and M-Theory

Lecture 3: From Open-String Quanta to Tachyons and Gravity

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Chapter 1

From Open-String Quanta to Tachyons and Gravity

Overview. We now quantize the normal modes found in the previous lecture and ask what particle states they produce. Each transverse mode is a harmonic oscillator, but its normalization is slightly unfamiliar: exciting mode n raises the light-front energy, and hence the mass-squared level, by n units. The first excited open-string level transforms exactly like the two transverse polarizations of a massless vector. Relativistic consistency therefore fixes the ground-state mass squared to be negative. That state is the bosonic-string tachyon, whose real significance is an unstable vacuum, not a practical faster-than-light particle. Finally, once strings interact, joining and splitting make closed strings unavoidable; the lowest closed-string excitation will be the graviton.

1.1 The oscillator hidden in each string mode

For one transverse coordinate X , the n th nonzero normal mode has energy and Lagrangian

$$E_n = \frac{1}{4} \dot{X}_n^2 + \frac{n^2}{4} X_n^2, \quad (1.1)$$

$$L_n = \frac{1}{4} \dot{X}_n^2 - \frac{n^2}{4} X_n^2. \quad (1.2)$$

The factors of $1/4$ are not an error. They record the normalization chosen when the string action was reduced to independent Fourier modes. The first step in constructing the Hamiltonian is therefore to compute the canonical momentum,

$$p_n = \frac{\partial L_n}{\partial \dot{X}_n} = \frac{1}{2} \dot{X}_n, \quad \dot{X}_n = 2p_n. \quad (1.3)$$

The Legendre transform gives

$$H_n = p_n \dot{X}_n - L_n = p_n^2 + \frac{n^2}{4} X_n^2. \quad (1.4)$$

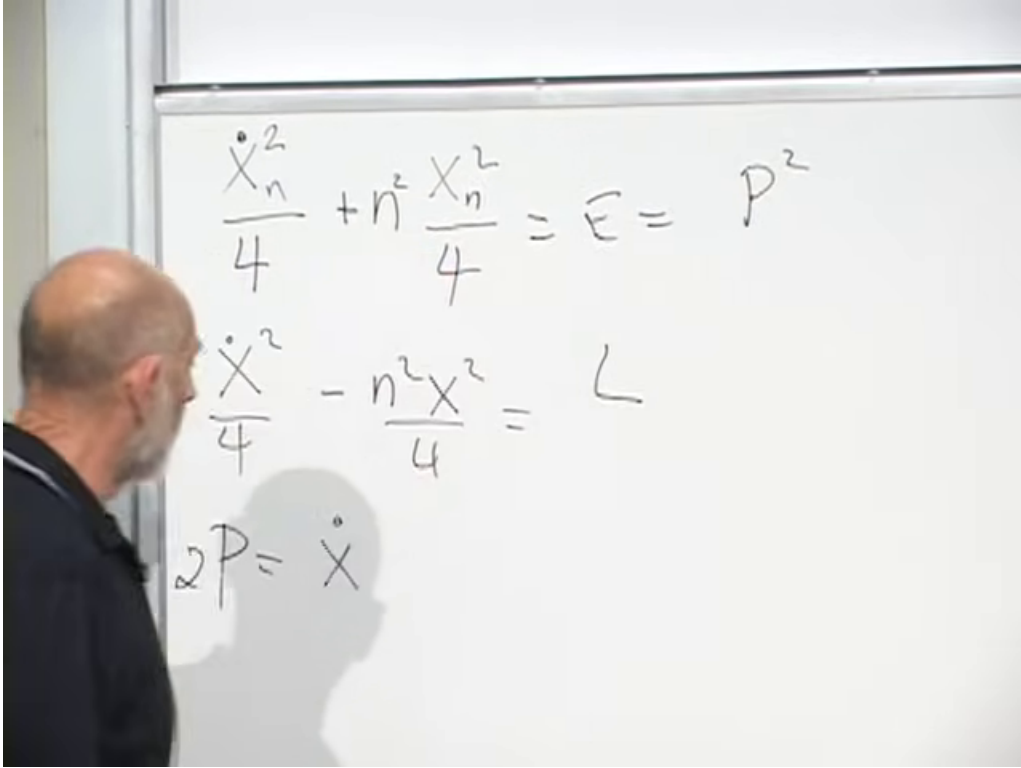


Figure 1.1: The mode energy, Lagrangian, canonical momentum, and Hamiltonian in the normalization inherited from the open string.

Lecture frame: 00:05:40.

To factorize H_n , combine X_n and p_n as

$$\frac{nX_n}{2} \pm ip_n. \quad (1.5)$$

Classically the two factors would commute, but quantum mechanically

$$[X_n, p_m] = i\delta_{nm} \quad (1.6)$$

in units with $\hbar = 1$. The cross terms do not cancel innocently. For a fixed mode,

$$\left[\frac{nX_n}{2} + ip_n, \frac{nX_n}{2} - ip_n \right] = n, \quad (1.7)$$

$$\left(\frac{nX_n}{2} - ip_n \right) \left(\frac{nX_n}{2} + ip_n \right) = H_n - \frac{n}{2}. \quad (1.8)$$

The image shows a whiteboard with handwritten mathematical expressions. The top expression is $= \left[p^2 + \frac{n^2 x^2}{4} \right] = H$. Below it, a commutator is calculated: $\left[\left(\frac{nx}{2} + ip \right) \left(\frac{nx}{2} - ip \right) \right] = \frac{n}{2} (-i)(i) = n$. Arrows indicate the expansion of the commutator into two terms.

Figure 1.2: The board calculation in which the canonical commutator supplies the factor of n .*Lecture frame: 00:10:30.*

The factor of n tells us how to normalize the ladder operators:

$$a_n \equiv \frac{\sqrt{n}}{2} X_n + \frac{i}{\sqrt{n}} p_n, \quad (1.9)$$

$$a_n^\dagger \equiv \frac{\sqrt{n}}{2} X_n - \frac{i}{\sqrt{n}} p_n. \quad (1.10)$$

They obey

$$[a_n, a_m^\dagger] = \delta_{nm}, \quad H_n = n \left(a_n^\dagger a_n + \frac{1}{2} \right). \quad (1.11)$$

Thus $a_n^\dagger$ adds one quantum of the n th oscillator, but that quantum carries n units of mode energy. The familiar oscillator spacing is not lost; its frequency is $\omega_n = n$.

Handwritten notes on a whiteboard:

$$[X, P] = i$$

$$a^- = \frac{\sqrt{n} X}{2} + \frac{i}{\sqrt{n}} P$$

$$a^+ = \frac{\sqrt{n} X}{2} - \frac{i}{\sqrt{n}} P$$

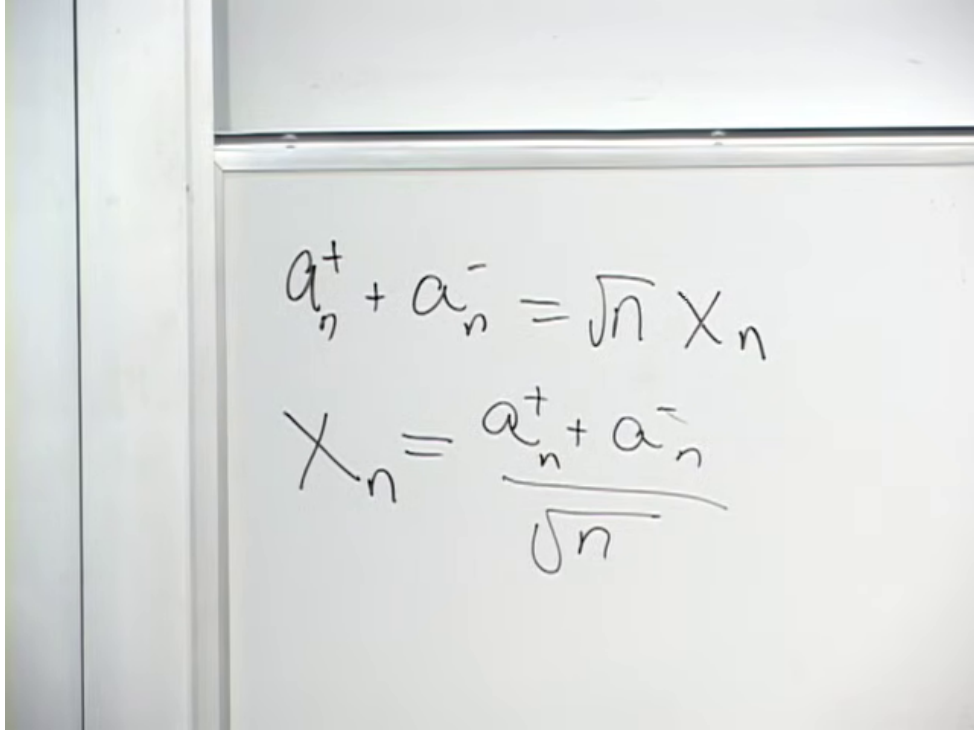
There is also a partially visible formula to the right: $\frac{1}{\sqrt{n}} \left(\frac{n X}{2} - \dots \right)$

Figure 1.3: Normalized annihilation and creation operators for the n th string oscillator.*Lecture frame: 00:12:00.*

1.1.1 Two transverse towers

In four spacetime dimensions, a string carrying very large momentum along the z -axis has two transverse coordinates, X and Y . We therefore need two independent oscillator towers. Let $a_n, a_n^\dagger$ act on the X modes and $b_n, b_n^\dagger$ on the Y modes. Solving (1.9)–(1.10) for the coordinates gives

$$X_n = \frac{a_n + a_n^\dagger}{\sqrt{n}}, \quad Y_n = \frac{b_n + b_n^\dagger}{\sqrt{n}}. \quad (1.12)$$



$$a_n^+ + a_n^- = \sqrt{n} X_n$$

$$X_n = \frac{a_n^+ + a_n^-}{\sqrt{n}}$$

Figure 1.4: Solving for a mode coordinate in terms of its creation and annihilation operators.

Lecture frame: 00:17:05.

The open-string coordinates were expanded in Neumann modes,

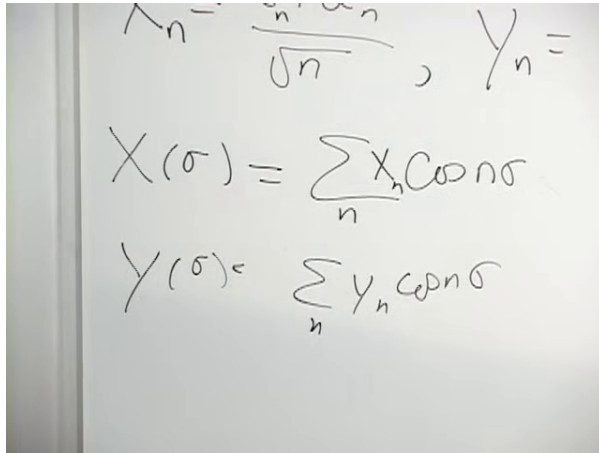
$$X(\sigma, \tau) = X_{\text{cm}}(\tau) + \sum_{n=1}^{\infty} X_n(\tau) \cos(n\sigma), \quad (1.13)$$

$$Y(\sigma, \tau) = Y_{\text{cm}}(\tau) + \sum_{n=1}^{\infty} Y_n(\tau) \cos(n\sigma). \quad (1.14)$$

Cosines appear because the endpoint slopes vanish. Substituting (1.12) turns the string itself into an operator:

$$X(\sigma, \tau) = X_{\text{cm}}(\tau) + \sum_{n=1}^{\infty} \frac{a_n(\tau) + a_n^\dagger(\tau)}{\sqrt{n}} \cos(n\sigma), \quad (1.15)$$

$$Y(\sigma, \tau) = Y_{\text{cm}}(\tau) + \sum_{n=1}^{\infty} \frac{b_n(\tau) + b_n^\dagger(\tau)}{\sqrt{n}} \cos(n\sigma). \quad (1.16)$$

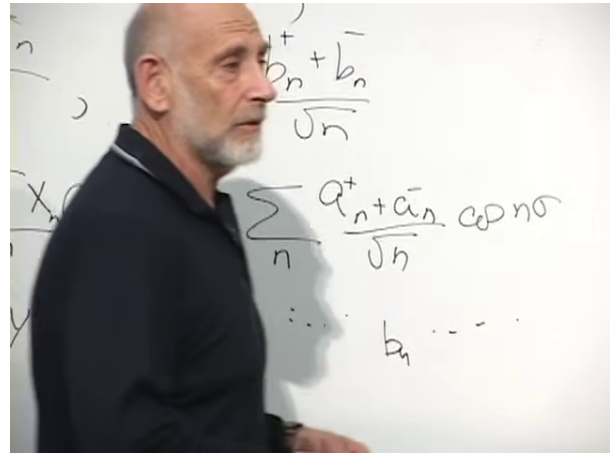


$$X_n = \frac{a_n + \bar{a}_n}{\sqrt{n}}, \quad Y_n = \frac{a_n - \bar{a}_n}{\sqrt{n}}$$

$$X(\sigma) = \sum_n X_n \cos n\sigma$$

$$Y(\sigma) = \sum_n Y_n \cos n\sigma$$

(a) Coordinate-mode expansions.



$$X_n = \frac{a_n + \bar{a}_n}{\sqrt{n}}$$

$$Y_n = \frac{a_n - \bar{a}_n}{\sqrt{n}}$$

$$X(\sigma) = \sum_n \frac{a_n + \bar{a}_n}{\sqrt{n}} \cos n\sigma$$

$$Y(\sigma) = \sum_n \frac{a_n - \bar{a}_n}{\sqrt{n}} \cos n\sigma$$

(b) Ladder operators substituted into the string coordinates.

Figure 1.5: The shape of the quantum open string is assembled from all of its transverse oscillator operators.

Lecture frame: 00:20:35 and 00:21:05.

Question from the audience. Is X_n also a function of σ ?^a

Response. No. The σ -dependence is entirely in $\cos(n\sigma)$. The coefficient X_n is the time-dependent amplitude of that spatial pattern, so $X_n = X_n(\tau)$.

^aLecture timestamp: 00:21:48.

Question from the audience. Why do the operators called raising and lowering seem to have their signs reversed?^a

Response. The visible sign of ip_n is not a naming rule. Compute the commutator with the Hamiltonian. The operator satisfying $[H_n, a_n^\dagger] = na_n^\dagger$ raises the energy; the one satisfying $[H_n, a_n] = -na_n$ lowers it.

^aLecture timestamp: 00:23:13.

The two transverse towers are a pedagogical four-dimensional picture. A bosonic string quantized consistently has twenty-six spacetime dimensions, so it has twenty-four transverse oscillator families rather than two. The logic below is unchanged: the number of first-level states must match the number of physical polarizations of a massless vector in the same dimension.

1.2 Massive spin and massless helicity

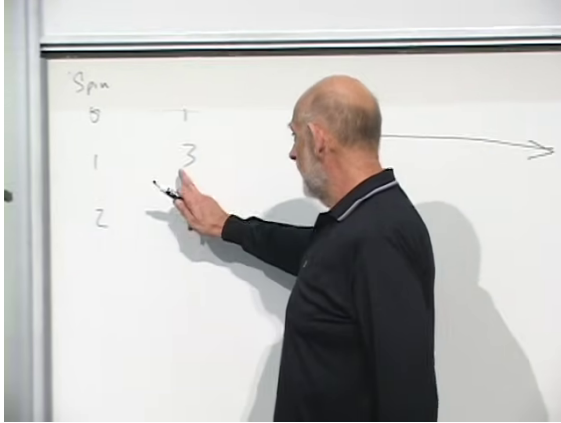
Before assigning spin to the string states, we need one sharp distinction. A massive particle of spin J has $2J + 1$ spin states,

$$m_J = -J, -J + 1, \dots, J - 1, J. \quad (1.17)$$

For example, a massive spin-one particle has three polarizations and a massive spin-two particle has five. We can see why by going to the rest frame. There every spatial rotation is available. Rotate

the spin axis, boost back, and all $2J + 1$ components belong to the same relativistic particle.

A massless particle has no rest frame. Its invariant spin label is helicity, the angular momentum along its direction of motion. A photon has only $h = +1$ and $h = -1$; a graviton has only $h = +2$ and $h = -2$. A scalar has one $h = 0$ state. The lecture uses reflection symmetry to pair the two helicities. In a chiral theory a single massless irreducible representation may carry only one helicity, so the pairing is an additional symmetry statement rather than a consequence of masslessness alone.¹



(a) The spin-state count begun on the board.

Spin	Massive states	Massless helicities
0	1	0
1	3	+1, -1
2	5	+2, -2

Figure 1.6: Mass removes the rest frame and with it the intermediate spin projections.

Lecture frame: 00:27:40.

1.2.1 Linear and circular photon polarization

Take a photon moving along z . Its two transverse linear polarizations are $|x\rangle$ and $|y\rangle$. They are not helicity eigenstates. The circular states are

$$|R\rangle = \frac{1}{\sqrt{2}} (|x\rangle + i|y\rangle), \quad (1.18)$$

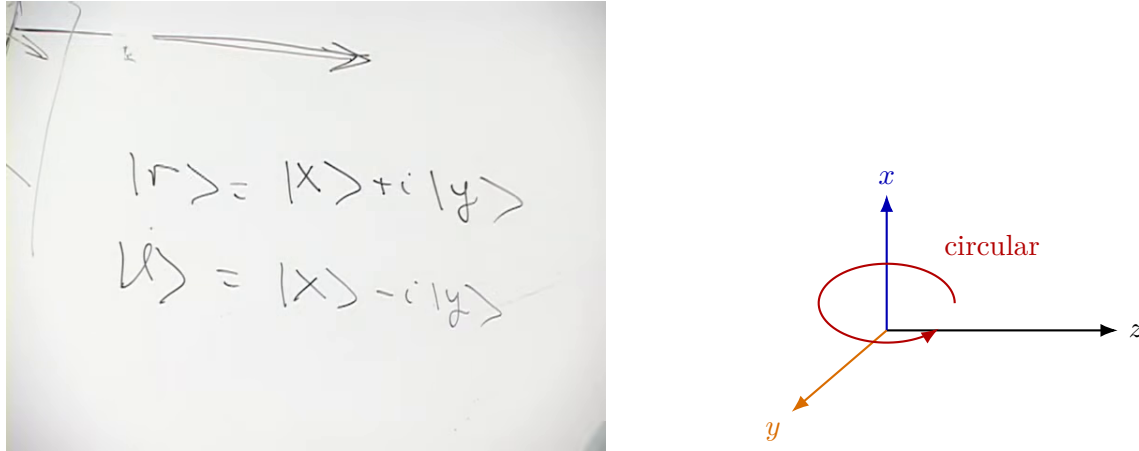
$$|L\rangle = \frac{1}{\sqrt{2}} (|x\rangle - i|y\rangle), \quad (1.19)$$

with helicities $+1$ and -1 , up to the convention used to name right and left. A linearly polarized photon is a superposition of those two outcomes. Its average helicity can vanish, but a helicity measurement never finds a spin-zero photon. Likewise,

$$\frac{1}{\sqrt{2}} (|x\rangle + |y\rangle) \quad (1.20)$$

is simply linear polarization at forty-five degrees.

¹Editorial clarification: The modern little-group classification permits one-helicity massless representations. Parity or an appropriate CPT-complete field content supplies the opposite helicity in the familiar photon and graviton examples.



(a) Circular polarization states on the board.

Figure 1.7: Linear polarization supplies a transverse basis; circular combinations diagonalize helicity.

Lecture frame: 00:37:45.

Question from the audience. Is this like the separation of spin states in a Stern–Gerlach experiment?^a

Response. The quantum-superposition idea is the same, but for the photon the direct language is polarization. A circular-polarization measurement distinguishes the two helicities; there is no longitudinal spin-zero photon in the beam.

^aLecture timestamp: 00:37:40.

The graviton repeats the pattern at spin two. For spin bookkeeping one may write schematically

$$|+2\rangle \sim |+1\rangle \otimes |+1\rangle, \quad |-2\rangle \sim |-1\rangle \otimes |-1\rangle. \quad (1.21)$$

This does not say that a graviton is literally made from two photons. It says that two equal helicities add to the correct spin-two helicity.

Question from the audience. Does the two-photon analogy extend to other quantum numbers?^a

Response. Not as a constituent model. There are powerful formulas in which gravity amplitudes look like products, or “squares,” of gauge-theory amplitudes, but the graviton is still its own particle. The analogy is most trustworthy here as a statement about helicity.^b

^aLecture timestamp: 00:41:05.

^bEditorial clarification: Modern Kawai–Lewellen–Tye and double-copy relations make the amplitude statement precise in suitable theories; they do not turn a graviton into a bound state of two photons.

Question from the audience. A photon can have any momentum, so does it not have a continuous spectrum rather than the isolated spectrum required of a particle?^a

Response. The continuous variable is the center-of-mass momentum. Hold that momentum fixed and inspect the internal invariant mass. A particle is identified by an isolated mass level; varying its overall momentum then produces the usual continuous dispersion relation.

^aLecture timestamp: 00:42:17.

1.3 The lowest open-string levels

We can now return to the string. In the large-momentum description, the finite internal light-front energy is proportional to invariant mass squared. Suppressing the overall string scale, write the spectrum as

$$M^2 = m_0^2 + \sum_{n=1}^{\infty} n \left(N_n^{(a)} + N_n^{(b)} \right), \quad (1.22)$$

$$N_n^{(a)} = a_n^\dagger a_n, \quad N_n^{(b)} = b_n^\dagger b_n.$$

The oscillator ground state obeys

$$a_n |0\rangle = 0, \quad b_n |0\rangle = 0, \quad n = 1, 2, \dots \quad (1.23)$$

It is not an empty absence of string. It is the least excited quantum string, and its coordinates still fluctuate. Let its mass squared be m_0^2 , not yet determined.

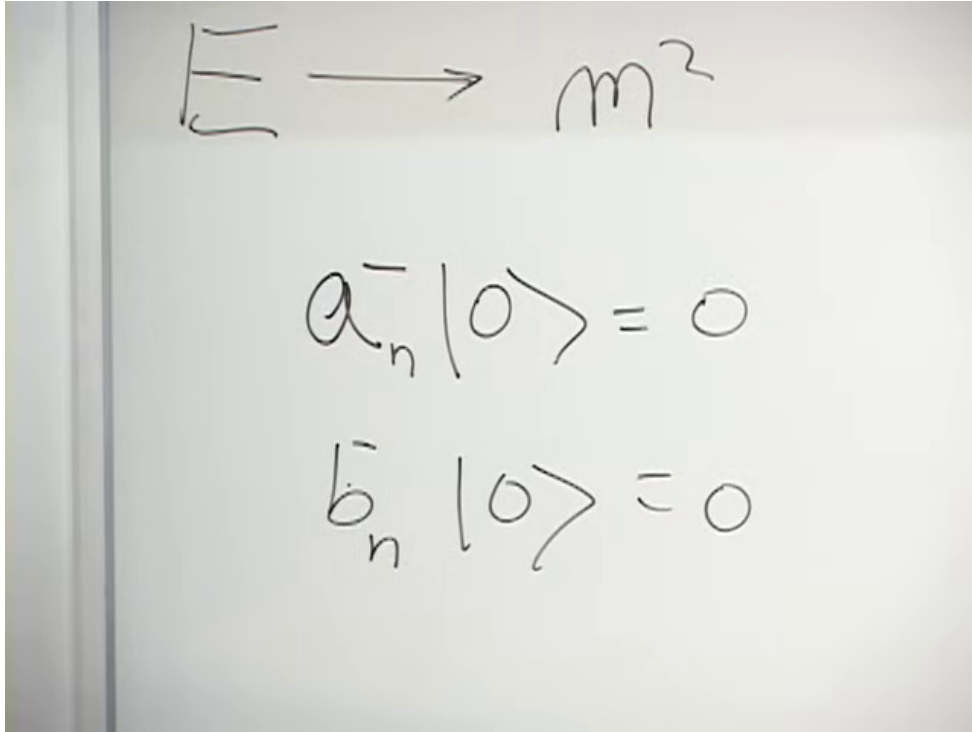


Figure 1.8: The open-string ground state and the annihilation conditions that define it.

Lecture frame: 00:45:20.

The cheapest excitations use the $n = 1$ creation operators:

$$a_1^\dagger |0\rangle, \quad b_1^\dagger |0\rangle, \quad M^2 = m_0^2 + 1. \quad (1.24)$$

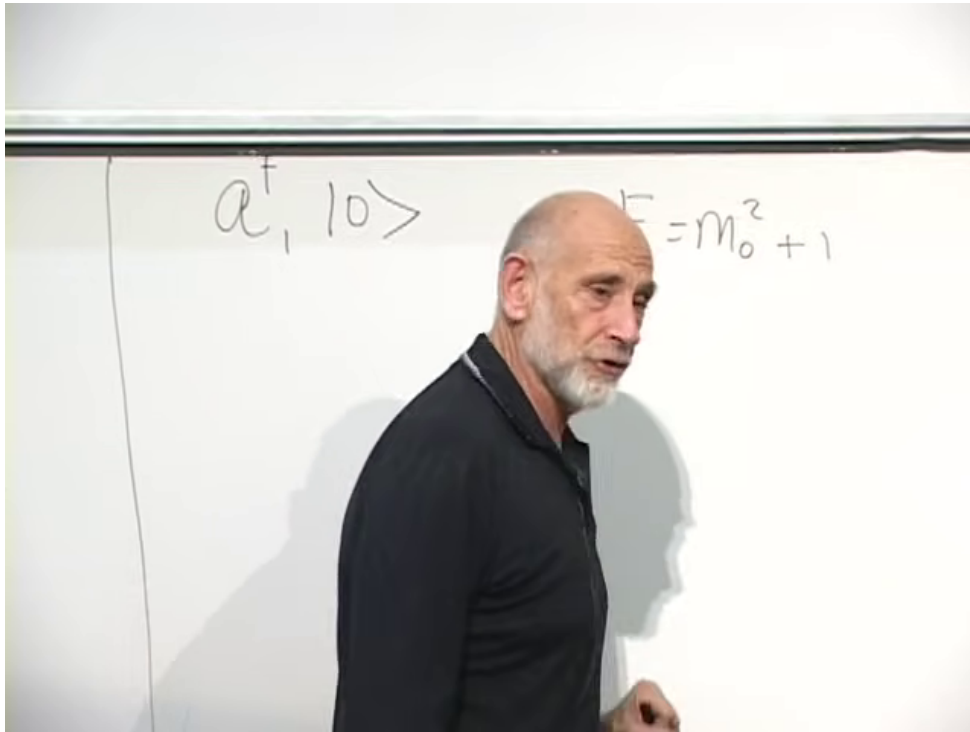


Figure 1.9: One of the two degenerate first-level states, with its mass-squared increment written on the board.

Lecture frame: 00:49:25.

The “one” in eqrefeq:l03-first-level is not one joule, one electron volt, or one Planck mass. We have suppressed the dimensionful string scale. For now it means one step in the oscillator’s mass-squared ladder; the missing units will be restored in the next lecture.

Any linear combination of the two degenerate states is also a state at this level. In particular,

$$\frac{1}{\sqrt{2}} (a_1^\dagger \pm i b_1^\dagger) |0\rangle \quad (1.25)$$

is the string analogue of circular polarization. Under a rotation in the x - y plane, $a_1^\dagger$ and $b_1^\dagger$ transform as the two components of a transverse vector. The combinations in (1.25) therefore have helicity $+1$ and -1 .

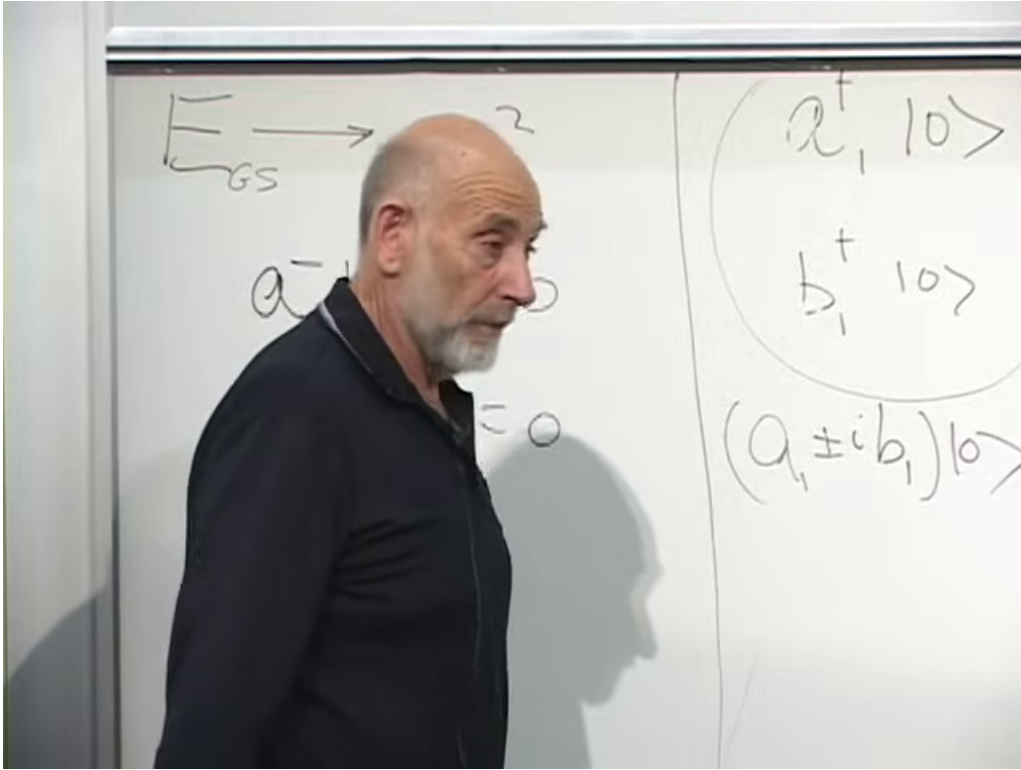


Figure 1.10: Ground-state conditions, the two first-level excitations, and their circular combinations collected on the board.

Lecture frame: 00:56:00.

There is no third degenerate state that could serve as a longitudinal polarization. If these two states had nonzero mass, a spin-one relativistic particle would require three polarizations. With exactly two, the state must be massless. Hence

$$m_0^2 + 1 = 0, \quad m_0^2 = -1. \quad (1.26)$$

The first level is photon-like: it is a massless gauge-vector state. It need not be the photon of electromagnetism; its eventual gauge identity depends on the string model. The ground state below it has negative mass squared. That is the bosonic open-string tachyon.²

Question from the audience. Why should we care that m_0^2 is negative?^a

Response. Because mass squared is the curvature of the quadratic field potential about the state we have called the vacuum. Negative curvature means that this configuration is not a stable ground state. The next calculation makes that statement precise.

^aLecture timestamp: 00:56:35.

²Editorial clarification: In the full critical bosonic string the normal-ordering intercept and Lorentz algebra are tied to $D = 26$. The lecture's two-transverse-direction calculation is the four-dimensional shadow of the same state-counting argument: $D - 2$ first-level oscillator states match the $D - 2$ physical polarizations of a massless vector.

1.4 What a tachyon really means

The name “tachyon” invites a misleading first thought. Start from the relativistic dispersion relation with $c = 1$,

$$E(p) = \sqrt{p^2 + m^2}. \quad (1.27)$$

For a wave packet, the group velocity is

$$v_g = \frac{dE}{dp}. \quad (1.28)$$

This reproduces familiar cases:

$$E_{\text{nr}}(p) = \frac{p^2}{2m}, \quad v_g = \frac{p}{m}, \quad (1.29)$$

$$E_\gamma(p) = |p|, \quad |v_g| = 1, \quad (1.30)$$

$$E_m(p) = \sqrt{p^2 + m^2}, \quad |v_g| = \frac{|p|}{\sqrt{p^2 + m^2}} < 1. \quad (1.31)$$

Handwritten equations on a whiteboard:

$$E(p) = \frac{p^2}{2m}$$

$$\frac{\partial E}{\partial p} = \frac{p}{m} = v$$

(a) Nonrelativistic group velocity.

Handwritten equations on a whiteboard:

$$E = \sqrt{p^2 + m^2}$$

$$\frac{\partial E}{\partial p} = \frac{p}{\sqrt{p^2 + m^2}}$$

(b) Relativistic dispersion and its derivative.

Figure 1.11: Reading propagation speed from the slope of $E(p)$.

Lecture frame: 01:01:50 and 01:03:15.

If $m^2 = -\mu^2$, the formal expression becomes

$$E(p) = \sqrt{p^2 - \mu^2}, \quad v_g = \frac{p}{\sqrt{p^2 - \mu^2}}, \quad (1.32)$$

which has $|v_g| > 1$ whenever the square root is real. But this is not the right physical interpretation of the bosonic-string ground state. The trouble appears already at low momentum, where the frequency is imaginary.

1.4.1 Negative curvature of the field potential

Consider a real scalar field in one spatial dimension,

$$\left(\partial_t^2 - \partial_x^2 + m^2\right)\phi = 0. \quad (1.33)$$

For a plane wave $\phi \propto e^{-i\omega t + ikx}$,

$$\omega^2 = k^2 + m^2. \quad (1.34)$$

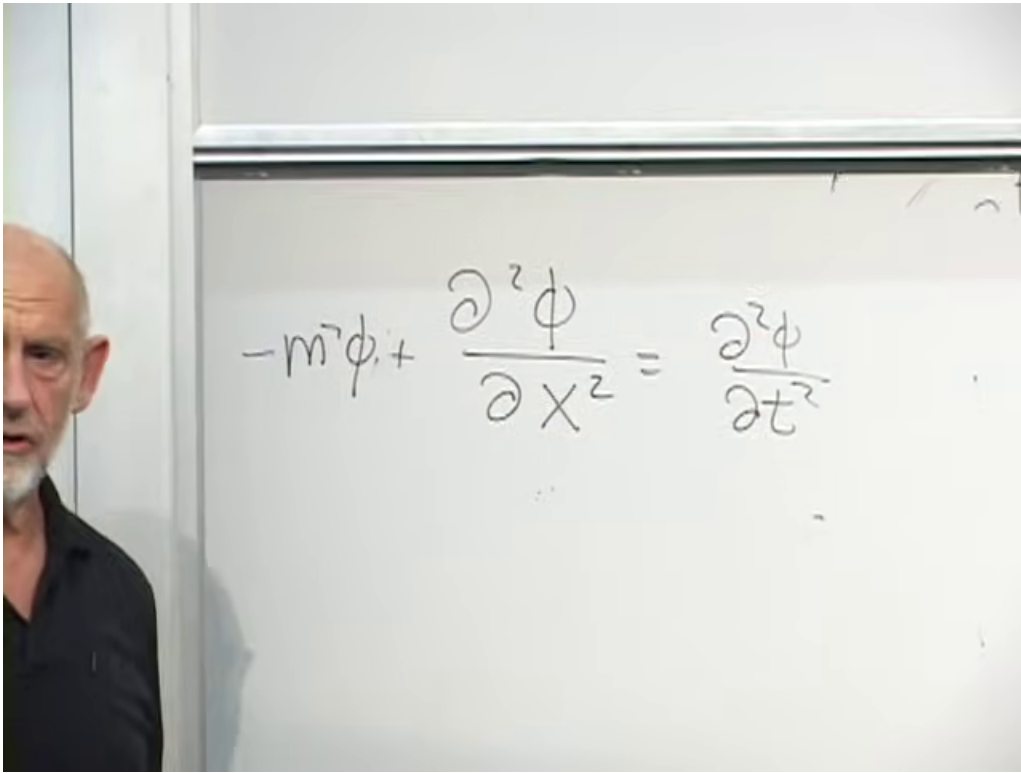


Figure 1.12: The Klein–Gordon equation used to expose the instability behind negative mass squared.

Lecture frame: 01:06:40.

The conserved quadratic energy is

$$\mathcal{E} = \frac{1}{2} \int dx \left[\dot{\phi}^2 + (\partial_x \phi)^2 + m^2 \phi^2 \right]. \quad (1.35)$$

$$\frac{\partial^2 \phi}{\partial x^2} = \frac{\partial^2 \phi}{\partial t^2}$$

$$\frac{1}{2} \left(\frac{\partial \phi}{\partial t} \right)^2 + \frac{1}{2} \left(\frac{\partial \phi}{\partial x} \right)^2 + \frac{m^2 \phi^2}{2}$$

Figure 1.13: The scalar-field energy, with the mass term supplying the local curvature of the potential.

Lecture frame: 01:08:10.

For $m^2 > 0$, the local potential $V(\phi) = \frac{1}{2}m^2\phi^2$ is a bowl. The configuration $\phi = 0$ sits at its bottom. For $m^2 = -\mu^2$, the same point is the top of an inverted bowl. Equation (1.34) becomes

$$\omega^2 = k^2 - \mu^2. \quad (1.36)$$

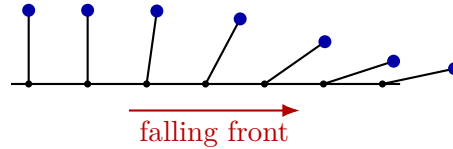
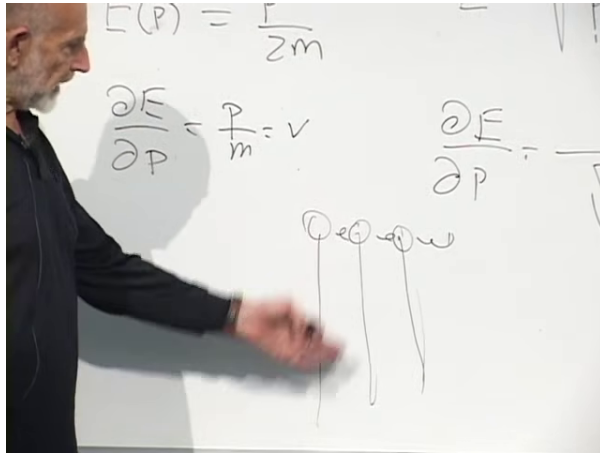
Modes with $k^2 < \mu^2$ have

$$\omega = \pm i\sqrt{\mu^2 - k^2}, \quad \phi_k(t) \propto e^{\pm\sqrt{\mu^2 - k^2}t}. \quad (1.37)$$

One branch grows exponentially. The alleged vacuum is therefore unstable: an arbitrarily small long-wavelength disturbance drives the field away from $\phi = 0$. That is what a tachyon in the spectrum means.

1.4.2 The upright-pendulum chain

A row of coupled pendulums makes the distinction visible. Put every pendulum upright. The uniform configuration exists, but it is balanced at a maximum of potential energy. Nudge one pendulum and it falls, pulling on its neighbors. A front of falling pendulums spreads through the chain at the finite signal speed fixed by the couplings.



(a) The coupled upright-pendulum model on the board.

Figure 1.14: Instability spreads causally through a system even though its linearized spectrum contains negative mass squared.

Lecture frame: 01:10:40.

Nothing in this mechanism permits a controlled signal to outrun the causal front. A tachyonic mode diagnoses a bad expansion point, not a stable superluminal messenger.³

Question from the audience. Is the massless state just found the graviton predicted by string theory?^a

Response. No. It is the two-polarization spin-one state of an open string, so it is gauge-boson-like. The graviton is a spin-two excitation of a closed string.

^aLecture timestamp: 01:14:27.

Question from the audience. Is an upside-down potential mathematically allowed, and where would the system finish after it falls?^a

Response. It is a perfectly legitimate local approximation, but by itself it has no ground state because the energy is unbounded below. A more complete potential may bend upward again and possess a lower minimum. The system then rolls or condenses toward that stable configuration.

^aLecture timestamp: 01:15:17.

The ordinary bosonic string has precisely this defect. Its higher oscillator states remain part of the spectrum, but they are excitations of an unstable background. Consistent superstring theories remove the open-string tachyon in their stable sectors while retaining massless and massive string excitations.

³Editorial clarification: For a local relativistic field equation the front velocity remains the causal speed. The formal group velocity of the real high- k branch does not turn the unstable vacuum into a faster-than-light communication channel.

Question from the audience. Do the other properties of the open-string state match a photon as well as its two polarizations do?^a

Response. The state behaves as a massless gauge boson. Its detailed charge assignments, radiation rules, and identity depend on the string model, so one should call it photon-like rather than identify it immediately with electromagnetism. Those interaction properties are developed later.

^aLecture timestamp: 01:17:07.

1.5 How strings interact

The free spectrum is only half the theory. Strings also join and split. It is useful first to remember that an excited free string can carry angular momentum. A straight rotating string turns about its center of mass, and its energy and angular momentum rise together along the familiar stringy trajectory.

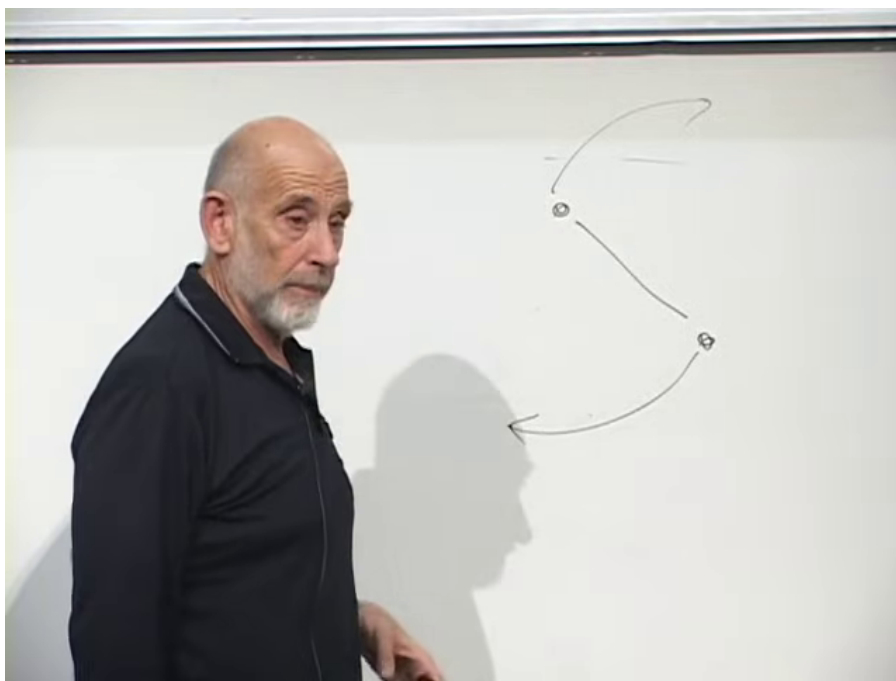


Figure 1.15: A rotating open string sketched about its center-of-mass point.

Lecture frame: 01:19:30.

Question from the audience. Can a free string rotate end over end about one fixed endpoint?^a

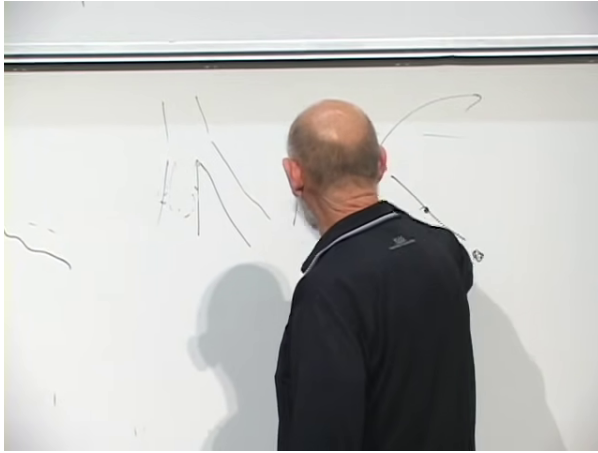
Response. Not without an anchor. A free isolated system rotates about its center of mass. If an endpoint is attached to another object or to a brane, the combined system can rotate about a different center, just as an electron and nucleus orbit their common center of mass while the much heavier nucleus appears nearly fixed.

^aLecture timestamp: 01:20:02.

1.5.1 The open-string vertex

For open strings the elementary interaction occurs when two endpoints meet at the same spacetime point. There is an amplitude g_s for them to join and form one string. Reversing the history gives an amplitude for one string to split into two. Schematically,

$$\text{open} + \text{open} \quad \rightleftharpoons \quad \text{open}. \quad (1.38)$$



(a) The joining history on the board.

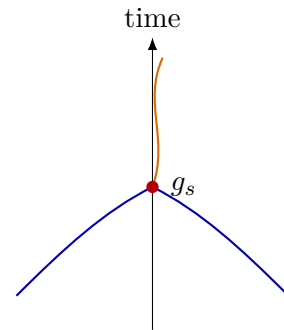


Figure 1.16: Endpoint joining is the string analogue of a local particle interaction vertex; time reversal gives splitting.

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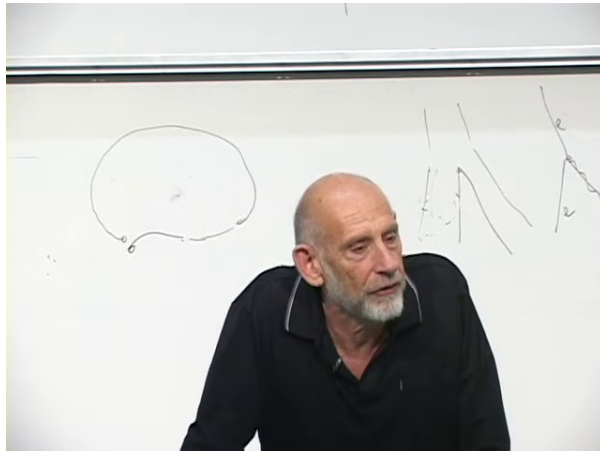
The coupling is an amplitude. For one specified elementary process the corresponding probability begins at order $|g_s|^2$, with phase-space and normalization factors supplied by the full calculation. The lecture moves between the colloquial words “probability” and “coupling”; keeping amplitude and probability distinct prevents a spurious square.

The interaction does not depend on whether the external string happens to be vibrating, rotating, or in some other oscillator state. Those states change the detailed matrix element, but the same local joining rule is used.

1.5.2 An open string can close itself

Suppose the two endpoints of one open string meet. The joining rule cannot know that they belong to the same string. They can fuse, producing a closed loop:

$$\text{open} \longrightarrow \text{closed}. \quad (1.39)$$



(a) The open string brings its endpoints together.

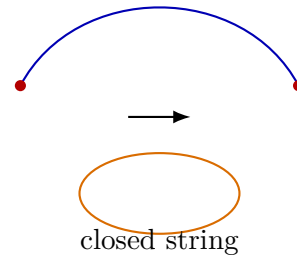


Figure 1.17: Interacting open strings inevitably generate closed strings.

Lecture frame: 01:27:20.

This is not an optional embellishment. Any theory of interacting open strings must contain closed-string states. The reverse process lets a closed string open into an open string when the allowed endpoints and boundary conditions are present.

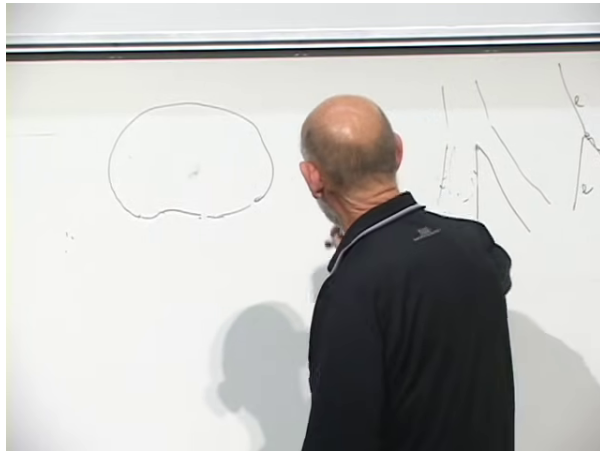
Question from the audience. When strings fuse or split, does their oscillator energy change, and does the theory determine how?^a

Response. Yes. Energy and momentum are conserved for the complete process, while excitation can be redistributed among the outgoing strings. The interaction vertex and the external wavefunctions determine the transition amplitude.

^aLecture timestamp: 01:28:05.

1.5.3 Closed strings can interact without opening

The implication is one-way. Interacting open strings require closed strings, but a theory may begin with only closed strings. Two closed loops can touch, reconnect, and become one closed loop; the reverse history splits one loop into two. No open endpoint need appear.



(a) Closed-string fusion on the board.

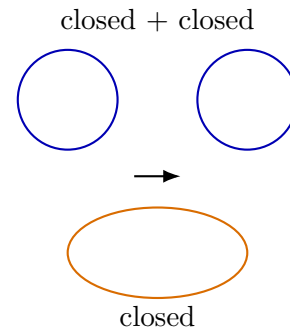


Figure 1.18: Closed strings possess their own joining and splitting interaction and need not pass through an open-string state.

Lecture frame: 01:29:05.

Question from the audience. If a theory starts with closed strings, must it also contain open strings?^a

Response. No. Closed strings can interact by reconnection while remaining closed throughout. Open strings require additional boundary data, such as allowed endpoints or branes.

^aLecture timestamp: 01:28:35.

1.6 Why gravity cannot be avoided

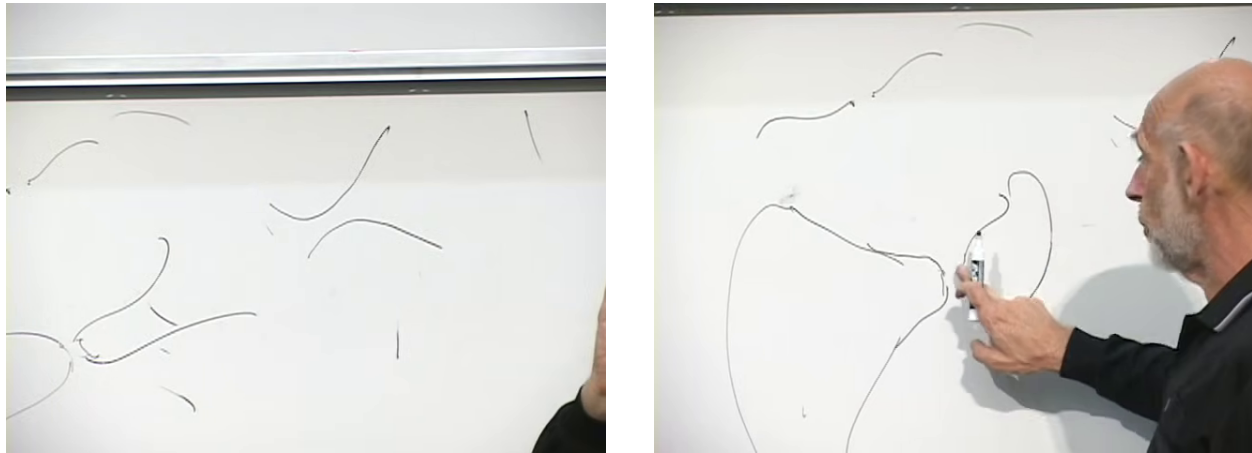
The closed-string oscillator calculation parallels the open-string one, but there are independent left-moving and right-moving excitations. Their lowest nontrivial matched level contains a massless spin-two state. That state has the helicities of a graviton. Consequently every consistent interacting closed-string theory contains gravity.

Question from the audience. Why does the lowest closed-string excitation have to be a graviton, just as the open-string level had to be photon-like? ^a

Response. The answer is the same kind of state counting, now with left- and right-moving oscillator vectors. Their tensor product contains a massless symmetric traceless spin-two polarization. The explicit closed-string spectrum will be worked out next; at this stage the point is the inevitability of that state.

^aLecture timestamp: 01:30:59.

Open and closed strings may also interact with one another. A closed loop can meet an open string and be absorbed, leaving a larger or differently excited open string. Conversely an open string can radiate a closed loop. At the particle level this is the emission or absorption of a graviton by matter.



(a) Local reconnection of string segments.

(b) A closed loop attaches to an open string.

Figure 1.19: The board pictures behind mixed open–closed interactions and the universal possibility of graviton exchange.

Lecture frame: 01:33:10 and 01:34:35.

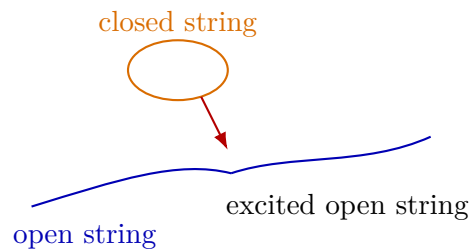


Figure 1.20: A clean spacetime-free schematic of closed-string absorption by an open string.

The graviton can couple to every string state because every string carries energy and momentum and the joining rules allow closed-string exchange. This is the string-theory origin of gravity’s universality.

Question from the audience. Can a closed string become an open string, or can it join an open string without violating the endpoint rule?^a

Response. In a theory containing both sectors, a closed string can attach to an open string and change its state. Whether a lone closed string may simply open depends on whether the theory supplies permissible endpoints. The local reconnection rule remains the same.

^aLecture timestamp: 01:31:20.

Question from the audience. Can closed loops become permanently linked or knotted?^a

Response. At weak coupling strings are mostly transparent: they can pass through one another unless the small reconnection amplitude acts. A link or knot can therefore persist for some time, but topology can change when an allowed string interaction occurs.

^aLecture timestamp: 01:35:10.

Question from the audience. Is the interaction probability independent of the state or speed of the string?^a

Response. The local coupling is the same parameter, but the total transition probability also depends on the wavefunctions and on how long the relevant pieces remain in contact. A fast encounter generally offers less interaction time than a slow one.

^aLecture timestamp: 01:36:09.

1.7 The scale and coupling still missing

Two pieces of bookkeeping have deliberately been postponed. First, X is a length while a_n is dimensionless, so the proportionality in (1.12) must contain a length scale. Second, raising the oscillator level cannot literally add a dimensionless number to a physical mass squared. Restoring the string length ℓ_s , or equivalently the Regge slope $\alpha' \sim \ell_s^2$, supplies the missing dimensions:

$$M^2 \sim \frac{1}{\alpha'}(N - a). \quad (1.40)$$

The precise convention and normalization will be fixed in the next lecture.

Question from the audience. What fixes the unit of string length and the mass added by one creation operator?^a

Response. A dimensionful constant has been suppressed. It is the string scale, usually written ℓ_s or α' . The oscillator operators are dimensionless; powers of that scale convert their level number into physical length, tension, and mass squared.

^aLecture timestamp: 01:37:03.

The second parameter is the string coupling g_s . In the simplest discussion it is one universal number controlling joining and splitting. In richer models, endpoints can carry different labels and strings can meet at junctions. A Y-shaped configuration is the string analogue of three flux tubes meeting; in the old hadronic picture it is associated with a three-quark baryon, whereas a single open segment resembles a quark–antiquark meson.

Question from the audience. Can different string endpoints have different joining probabilities?^a

Response. The simplest model uses one coupling. More elaborate theories attach charges or boundary labels to endpoints, and selection rules then decide which joins are allowed. The detailed construction is deferred, but the universal local joining idea survives.

^aLecture timestamp: 01:39:31.

Question from the audience. Can a string split lengthwise or form a Y-shaped junction?^a

Response. Some string systems do admit junctions. They are not the ordinary two-endpoint splitting process just drawn, but they occur when the theory has compatible string charges. The meson–baryon flux-tube analogy is a useful picture of the distinction.

^aLecture timestamp: 01:40:01.

Question from the audience. Does changing the number of spacetime dimensions change the interaction coupling?^a

Response. Dimensions change the spectrum, phase space, and effective couplings obtained after compactification. The fundamental string coupling still organizes the perturbative expansion, but its lower-dimensional appearance depends on the geometry and volume of the extra dimensions.

^aLecture timestamp: 01:42:25.

Question from the audience. Is gravity what makes a free string move in the first place?^a

Response. No force is required to maintain motion. Initial position and velocity specify a free solution; inertia carries it forward. Gravity or another force changes that motion. Asking what caused the initial conditions is a separate question, not part of the local equation of motion.

^aLecture timestamp: 01:43:17.

We have therefore reached three linked conclusions. Quantizing an open string produces a massless vector at its first excited level; the bosonic ground state below it is unstable; and interactions inevitably introduce closed strings, whose spectrum contains gravity. The next task is to restore the missing scale and perform the closed-string quantization explicitly.